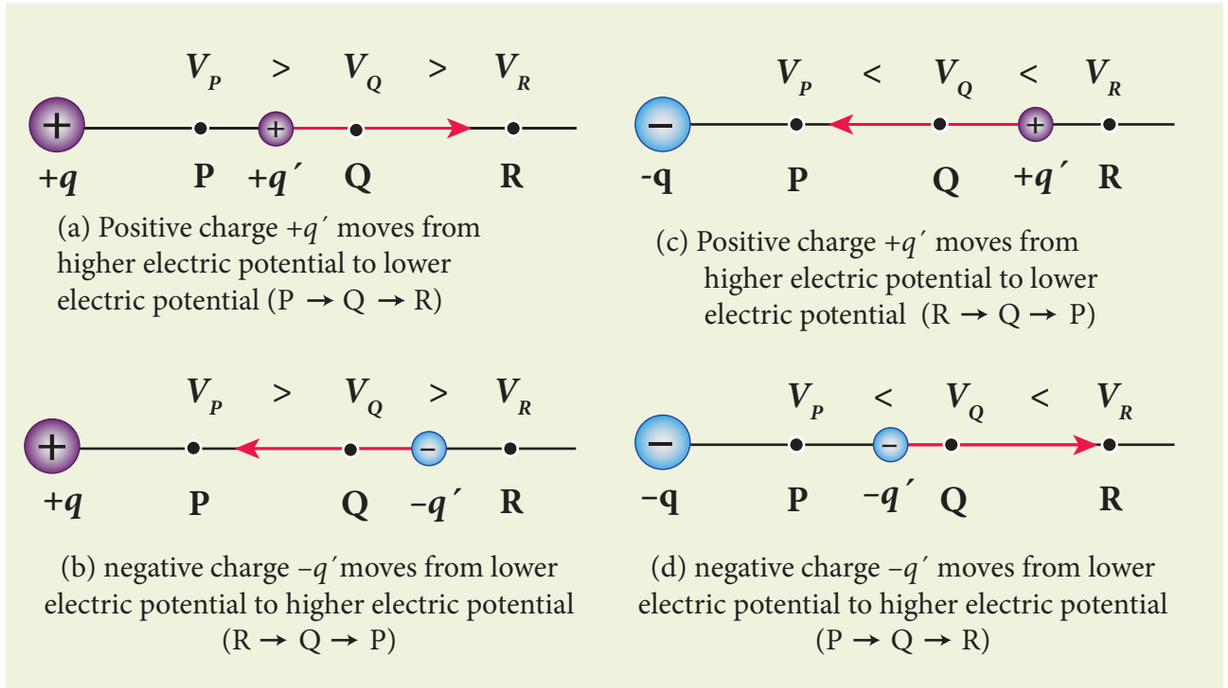


Figure 1.23 Motion of charges in terms of electric potential

gravitational potential (Figure 1.22). Similarly a positive charge moves from a point of higher electrostatic potential to a point of lower electrostatic potential. However a negative charge moves from lower electrostatic potential to higher electrostatic potential. This comparison is shown in Figure 1.23.

(iii) The electric potential at a point P due to a collection of charges $q_1, q_2, q_3 \ldots . q_n$ is equal to sum of the electric potentials due to individual charges.

$$V_{tot} = \frac{kq_1}{r_1} + \frac{kq_2}{r_2} + \frac{kq_3}{r_3} + \ldots$$
$$\ldots . + \frac{kq_n}{r_n} = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^{n} \frac{q_i}{r_i} \qquad (1.34)$$

where $r_1, r_2, r_3 \ldots \ldots . r_n$ are the distances of q_1, q_2, q_3 ….. q_n respectively from P (Figure 1.24).

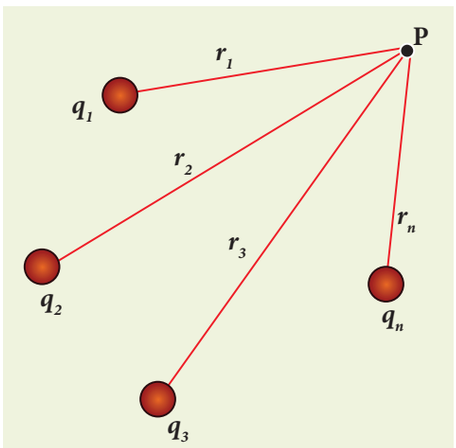


Figure 1.24 Electrostatic potential due to collection of charges

EXAMPLE 1.12

(a) Calculate the electric potential at points P and Q as shown in the figure below.

(b) Suppose the charge +9 μC is replaced by −9 μC find the electrostatic potentials at points P and Q

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